Part 1 — Evaluate — Key

1. $\arcsin\left(\frac{\sqrt{2}}{2}\right) = \frac{\pi}{4}$

2. $\arccos\left(-\frac{1}{2}\right) = \frac{2\pi}{3}$

3. $\arctan\left(\frac{\sqrt{3}}{3}\right) = \frac{\pi}{6}$

4. $\sin^{-1}(0) = 0$

5. $\cos^{-1}(1) = 0$

6. $\tan^{-1}\left(-\frac{\sqrt{3}}{3}\right) = -\frac{\pi}{6}$

Part 2 — Domain and Range — Key

7.

Function	Domain	Range
$y = \arcsin x$	$[-1, 1]$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
$y = \arccos x$	$[-1, 1]$	$[0, \pi]$
$y = \arctan x$	$(-\infty, \infty)$	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

Part 3 — AP-Style Multiple Choice — Key

8. (D) $\frac{3\pi}{4}$

$\cos\left(\frac{3\pi}{4}\right) = -\frac{\sqrt{2}}{2}$ and $\frac{3\pi}{4} \in [0, \pi]$ (restricted domain of cosine).

9. (C) The range of f is $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

The range of arctan is the open interval $(-\frac{\pi}{2}, \frac{\pi}{2})$. (A) is false — range is not all reals. (B) is false — domain of arctan is all reals. (D) is false — arctan is an odd function, so $f(-x) = -f(x)$ is true.

Extension optional

Extension — Composition of Trig and Inverse Trig — Key

10. Let $\theta = \arctan x$ with $x > 0$, so $\tan\theta = x$. Right triangle: opposite $= x$, adjacent $= 1$, hypotenuse $= \sqrt{x^2+1}$.

$\sin(\arctan x) = \frac{x}{\sqrt{x^2+1}}$

Preview Lesson 3.10: $\cos\theta = \frac{1}{2}$ has **two** solutions on $[0, 2\pi)$: $\theta = \frac{\pi}{3}$ and $\theta = \frac{5\pi}{3}$. Using arccos: $\arccos\left(\frac{1}{2}\right) = \frac{\pi}{3}$ gives one solution; the other is found by symmetry.